EXERCISE 1

Eight Spiral and Centre Portraits from Complex Eigen-data

Original statement · probls_ch4-ii_2627.pdf

Complex Eigenvalues • Spirals and Centres • Eight Portraits

STATEMENT

Draw the phase portrait for the equation $d\mathbf{x}/dt = A\mathbf{x}$, when the eigenvalues ($\lambda_\pm$) and eigenvectors ($\boldsymbol{\eta}_\pm$) of A are as given. The sign of $\dot{x}$ is indicated when $x = 0$ and $y > 0$.

- (i) $\lambda_\pm = 1 \pm 3i$ with $\boldsymbol{\eta}_\pm = (1,\ 2 \mp i)$, $\dot{x} < 0$;
- (ii) $\lambda_\pm = \pm 3i$ with $\boldsymbol{\eta}_\pm = (1 \pm 2i,\ 1 \mp 3i)$, $\dot{x} < 0$;
- (iii) $\lambda_\pm = -2 \pm i$ with $\boldsymbol{\eta}_\pm = (1 \mp i,\ 3 \pm i)$, $\dot{x} > 0$;
- (iv) $\lambda_\pm = -1 \pm i$ with $\boldsymbol{\eta}_\pm = (1,\ \pm i)$, $\dot{x} > 0$;
- (v) $\lambda_\pm = 2 \pm 2i$ with $\boldsymbol{\eta}_\pm = (\pm 3i,\ 5 \mp 4i)$, $\dot{x} < 0$;
- (vi) $\lambda_\pm = 5 \pm 3i$ with $\boldsymbol{\eta}_\pm = (2 \pm 5i,\ \mp i)$, $\dot{x} < 0$;
- (vii) $\lambda_\pm = \pm 7i$ with $\boldsymbol{\eta}_\pm = (1 \pm i,\ -1 \pm 2i)$, $\dot{x} > 0$;
- (viii) $\lambda_\pm = -13 \pm 17i$ with $\boldsymbol{\eta}_\pm = (\pm 6i - 8,\ 4 \mp 5i)$, $\dot{x} > 0$.

PHASE 1: SETUP AND CLASSIFICATION

In each item $\lambda_\pm = \rho \pm i\omega$ with $\omega > 0$ and $\boldsymbol{\eta}_+ = \mathbf{u} + i\mathbf{w}$ is the eigenvector of $\lambda_+ = \rho + i\omega$ (the second eigenvector $\boldsymbol{\eta}_- = \bar{\boldsymbol{\eta}}_+$ carries no new information). The type is fixed by ρ: $\rho < 0$ stable spiral, $\rho > 0$ unstable spiral, $\rho = 0$ centre (Topic 4 Sec. 7.6). The sense of rotation follows from the vector field at $(0,1)^T$: $\dot{\mathbf{x}} = A(0,1)^T = (a_{12}, a_{22})^T$, and $\dot{x} = a_{12} > 0$ means clockwise, $\dot{x} < 0$ counter-clockwise (Topic 4 Sec. 7.8). The sign of $\dot{x}$ given in the statement is therefore the sense of rotation:

Item	ρ	ω	Type	Given sign of $\dot{x}$ at $(0,1)$	Sense
(i)	1	3	unstable spiral	< 0	counter-clockwise
(ii)	0	3	centre	< 0	counter-clockwise
(iii)	-2	1	stable spiral	> 0	clockwise
(iv)	-1	1	stable spiral	> 0	clockwise
(v)	2	2	unstable spiral	< 0	counter-clockwise
(vi)	5	3	unstable spiral	< 0	counter-clockwise

Item	ρ	ω	Type	Given sign of $\dot{x}$ at $(0,1)$	Sense
(vii)	0	7	centre	>0	clockwise
(viii)	-13	17	stable spiral	>0	clockwise

PHASE 2: FORMULATION

For each item the eigenvector gives $\mathbf{u}=\operatorname{Re}\boldsymbol{\eta}_+$ and $\mathbf{w}=\operatorname{Im}\boldsymbol{\eta}_+$, hence $Q=[\mathbf{u}\mid\mathbf{w}]$ and the canonical form $J=\begin{pmatrix}\rho & \omega\\ -\omega & \rho\end{pmatrix}$ (rotation clockwise in the $\mathbf{z}$-plane). The matrix $A=QJQ^{-1}$ is rebuilt, the given sign of $\dot{x}$ is checked against a_{12}, and the portrait is drawn in the $\mathbf{z}$-plane and in the (x,y)-plane ($\mathbf{x}=Q\mathbf{z}$).

METHOD OF THE CLASS NOTES [NOTES, P. 10]

Case 2 of the notes: with $\mathbf{v}_\pm=\mathbf{v}_1\pm i\mathbf{v}_2$ (here $\mathbf{u}=\mathbf{v}_1$, $\mathbf{w}=\mathbf{v}_2$) and $Q=(\mathbf{v}_1\mid\mathbf{v}_2)$, $\mathbf{z}=Q^{-1}\mathbf{x}$ satisfies $\dot{\mathbf{z}}=\begin{pmatrix}\rho & \omega\\ -\omega & \rho\end{pmatrix}\mathbf{z}$. In polar form $r=r(0)e^{\rho t}$ and $\theta=\theta(0)-\omega t$: clockwise in the $\mathbf{z}$-plane, a stable spiral for $\rho<0$, an unstable spiral for $\rho>0$ and a centre for $\rho=0$. The sense of rotation in the (x,y)-plane is read, as in the example of the notes, from the sign of $\dot{x}$ on the positive y-axis.

PHASE 3: DERIVATION OF THE EIGHT PORTRAITS

(i) $\lambda_\pm=1\pm3i$ with $\boldsymbol{\eta}_\pm=(1,\ 2\mp i)$, $\dot{x}<0$

Split the eigenvector of $\lambda_+=1+3i$ into real and imaginary parts, $\boldsymbol{\eta}_+=\begin{pmatrix}1\\ 2-i\end{pmatrix}=\mathbf{u}+i\mathbf{w}$, so $\rho=1,\omega=3$ and

$$\mathbf{u}=\begin{pmatrix}1\\2\end{pmatrix},\qquad \mathbf{w}=\begin{pmatrix}0\\-1\end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u}=\rho\mathbf{u}-\omega\mathbf{w}$, $A\mathbf{w}=\omega\mathbf{u}+\rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ=QJ$ with $Q=[\mathbf{u}\mid\mathbf{w}]$, so $A=QJQ^{-1}$:

$$Q=\begin{pmatrix}1 & 0\\ 2 & -1\end{pmatrix},\quad J=\begin{pmatrix}1 & 3\\ -3 & 1\end{pmatrix},\quad \det Q=-1,\quad Q^{-1}=\begin{pmatrix}1 & 0\\ 2 & -1\end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 1\cdot 1 + 0\cdot(-3) = 1\\
(QJ)_{12} &= 1\cdot 3 + 0\cdot 1 = 3\\
(QJ)_{21} &= 2\cdot 1 + (-1)\cdot(-3) = 5\\
(QJ)_{22} &= 2\cdot 3 + (-1)\cdot 1 = 5
\end{aligned}
$$

$$
QJ = \begin{pmatrix} 1 & 3 \\ 5 & 5 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= 1\cdot 1 + 3\cdot 2 = 7\\
(A)_{12} &= 1\cdot 0 + 3\cdot(-1) = -3\\
(A)_{21} &= 5\cdot 1 + 5\cdot 2 = 15\\
(A)_{22} &= 5\cdot 0 + 5\cdot(-1) = -5
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} 7 & -3 \\ 15 & -5 \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = -3$, which is negative, in agreement with the statement ($\dot{x} < 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 1\begin{pmatrix} 1 \\ 2 \end{pmatrix} - 3\begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 3\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 1\begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}
$$

$\det Q = (1)(-1) - (0)(2) = -1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-1}\begin{pmatrix} -1 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} 1 & 3 \\ 5 & 5 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= 1 \cdot 1 + 0 \cdot 5 = 1 \\ (Q^{-1}AQ)_{12} &= 1 \cdot 3 + 0 \cdot 5 = 3 \\ (Q^{-1}AQ)_{21} &= 2 \cdot 1 + (-1) \cdot 5 = -3 \\ (Q^{-1}AQ)_{22} &= 2 \cdot 3 + (-1) \cdot 5 = 1 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 1 & 3 \\ -3 & 1 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = z_1 + 3z_2, \qquad \dot{z}_2 = -3z_1 + z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot r - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0 \cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0 \cos\omega t + \sin\theta_0 \sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0 \cos\omega t - \cos\theta_0 \sin\omega t$:

$$z_1(t) = e^{1t}\left(c_1 \cos 3t + c_2 \sin 3t\right), \qquad z_2(t) = e^{1t}\left(c_2 \cos 3t - c_1 \sin 3t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1 \sin\omega t + c_2 \cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2 \sin\omega t - c_1 \cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

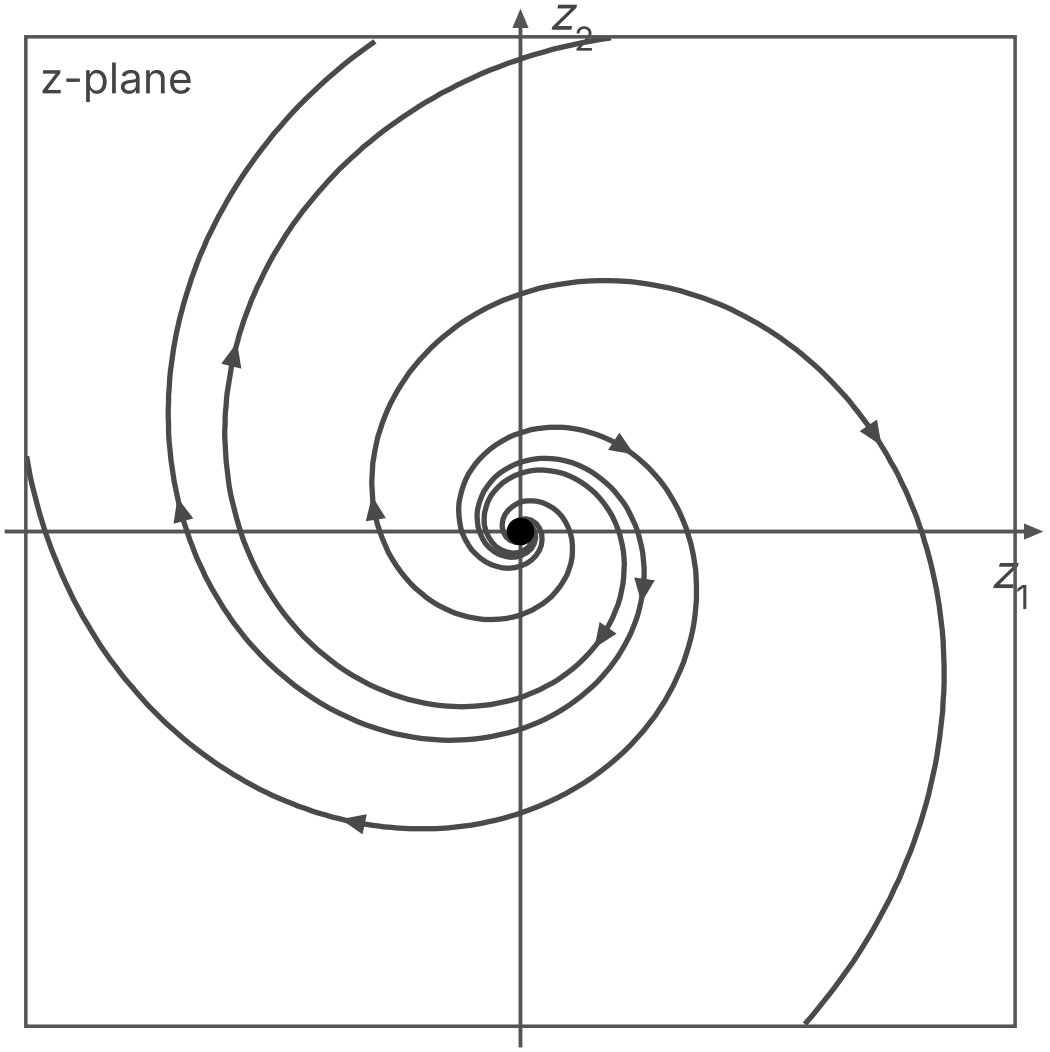
$$\mathbf{x}(t) = e^{1t}\left[\left(c_1 \cos 3t + c_2 \sin 3t\right)\begin{pmatrix}1\\2\end{pmatrix} + \left(c_2 \cos 3t - c_1 \sin 3t\right)\begin{pmatrix}0\\-1\end{pmatrix}\right]$$

$$x(t) = c_1 e^t \cos 3t + c_2 e^t \sin 3t, \qquad y(t) = c_1 e^t \sin 3t + 2c_1 e^t \cos 3t + 2c_2 e^t \sin 3t - c_2 e^t \cos 3t$$

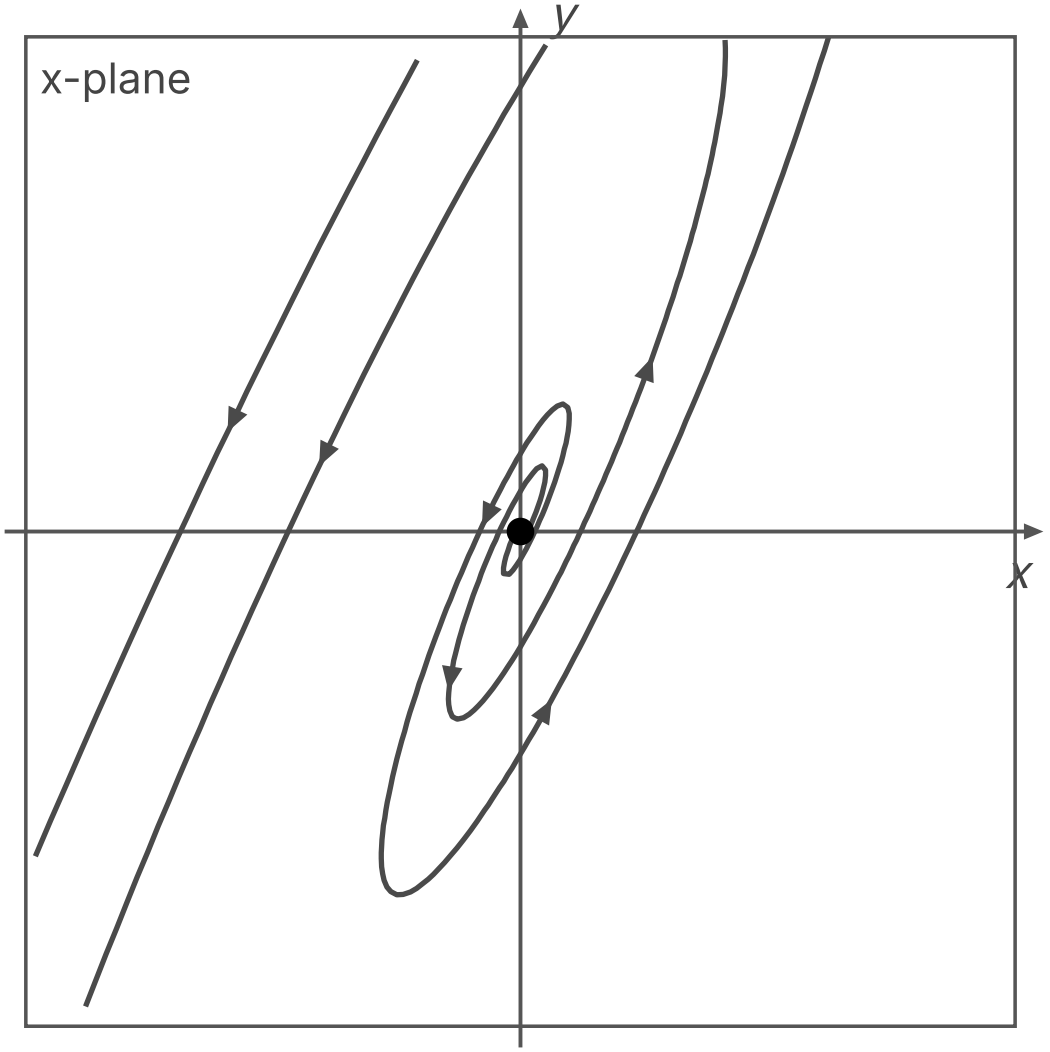
Phase portrait

$\rho = 1$ and $\omega = 3$: the origin is a **unstable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{1t}$ and $\theta = \theta_0 - 3t$: the motion is always **clockwise** ($\dot\theta = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{\frac{2\pi}{3}}$ (growing).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}-3\\-5\end{pmatrix}$, so $\dot x = -3$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: $\det Q = -1$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (unstable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (unstable spiral), counter-clockwise.

(ii) $\lambda_{\pm} = \pm 3i$ with $\boldsymbol{\eta}_{\pm} = (1 \pm 2i,\ 1 \mp 3i)$, $\dot{x} < 0$

Split the eigenvector of $\lambda_{+} = 3i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 1 + 2i \\ 1 - 3i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = 0$, $\omega = 3$ and

$$\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 1 & 2 \\ 1 & -3 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}, \quad \det Q = -5, \quad Q^{-1} = \begin{pmatrix} \frac{3}{5} & \frac{2}{5} \\ \frac{1}{5} & -\frac{1}{5} \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 1\cdot 0 + 2\cdot(-3) = -6\\
(QJ)_{12} &= 1\cdot 3 + 2\cdot 0 = 3\\
(QJ)_{21} &= 1\cdot 0 + (-3)\cdot(-3) = 9\\
(QJ)_{22} &= 1\cdot 3 + (-3)\cdot 0 = 3
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -6 & 3 \\ 9 & 3 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-6)\cdot\frac{3}{5} + 3\cdot\frac{1}{5} = -3\\
(A)_{12} &= (-6)\cdot\frac{2}{5} + 3\cdot(-\frac{1}{5}) = -3\\
(A)_{21} &= 9\cdot\frac{3}{5} + 3\cdot\frac{1}{5} = 6\\
(A)_{22} &= 9\cdot\frac{2}{5} + 3\cdot(-\frac{1}{5}) = 3
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} -3 & -3 \\ 6 & 3 \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = -3$, which is negative, in agreement with the statement ($\dot{x} < 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -6 \\ 9 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 0\begin{pmatrix} 1 \\ 1 \end{pmatrix} - 3\begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} -6 \\ 9 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix} + 0\begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 1 & 2 \\ 1 & -3 \end{pmatrix}
$$

$\det Q = (1)(-3) - (2)(1) = -5$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-5}\begin{pmatrix} -3 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{5} & \frac{2}{5} \\ \frac{1}{5} & -\frac{1}{5} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -6 & 3 \\ 9 & 3 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{3}{5}\cdot(-6) + \frac{2}{5}\cdot 9 = 0 \\ (Q^{-1}AQ)_{12} &= \frac{3}{5}\cdot 3 + \frac{2}{5}\cdot 3 = 3 \\ (Q^{-1}AQ)_{21} &= \frac{1}{5}\cdot(-6) + (-\frac{1}{5})\cdot 9 = -3 \\ (Q^{-1}AQ)_{22} &= \frac{1}{5}\cdot 3 + (-\frac{1}{5})\cdot 3 = 0 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = 3z_2, \qquad \dot{z}_2 = -3z_1$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0 \cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0 \cos\omega t + \sin\theta_0 \sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0 \cos\omega t - \cos\theta_0 \sin\omega t$:

$$z_1(t) = e^{0t}\left(c_1 \cos 3t + c_2 \sin 3t\right), \qquad z_2(t) = e^{0t}\left(c_2 \cos 3t - c_1 \sin 3t\right)$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1 \sin\omega t + c_2 \cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2 \sin\omega t - c_1 \cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

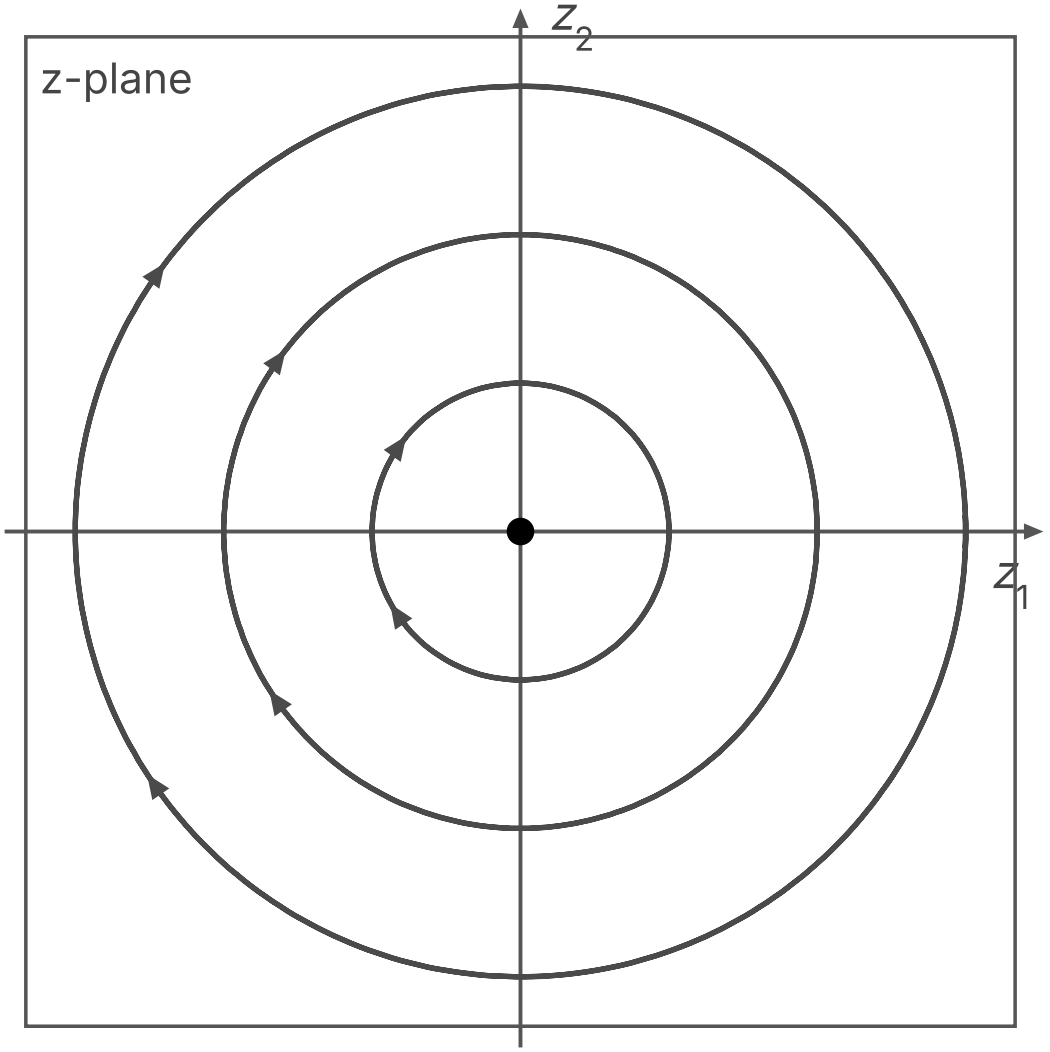
$$\mathbf{x}(t) = e^{0t}\left[\left(c_1 \cos 3t + c_2 \sin 3t\right)\begin{pmatrix}1\\1\end{pmatrix} + \left(c_2 \cos 3t - c_1 \sin 3t\right)\begin{pmatrix}2\\-3\end{pmatrix}\right]$$

$$x(t) = -2c_1 \sin 3t + c_1 \cos 3t + c_2 \sin 3t + 2c_2 \cos 3t, \qquad y(t) = 3c_1 \sin 3t + c_1 \cos 3t + c_2 \sin 3t - 3c_2 \cos 3t$$

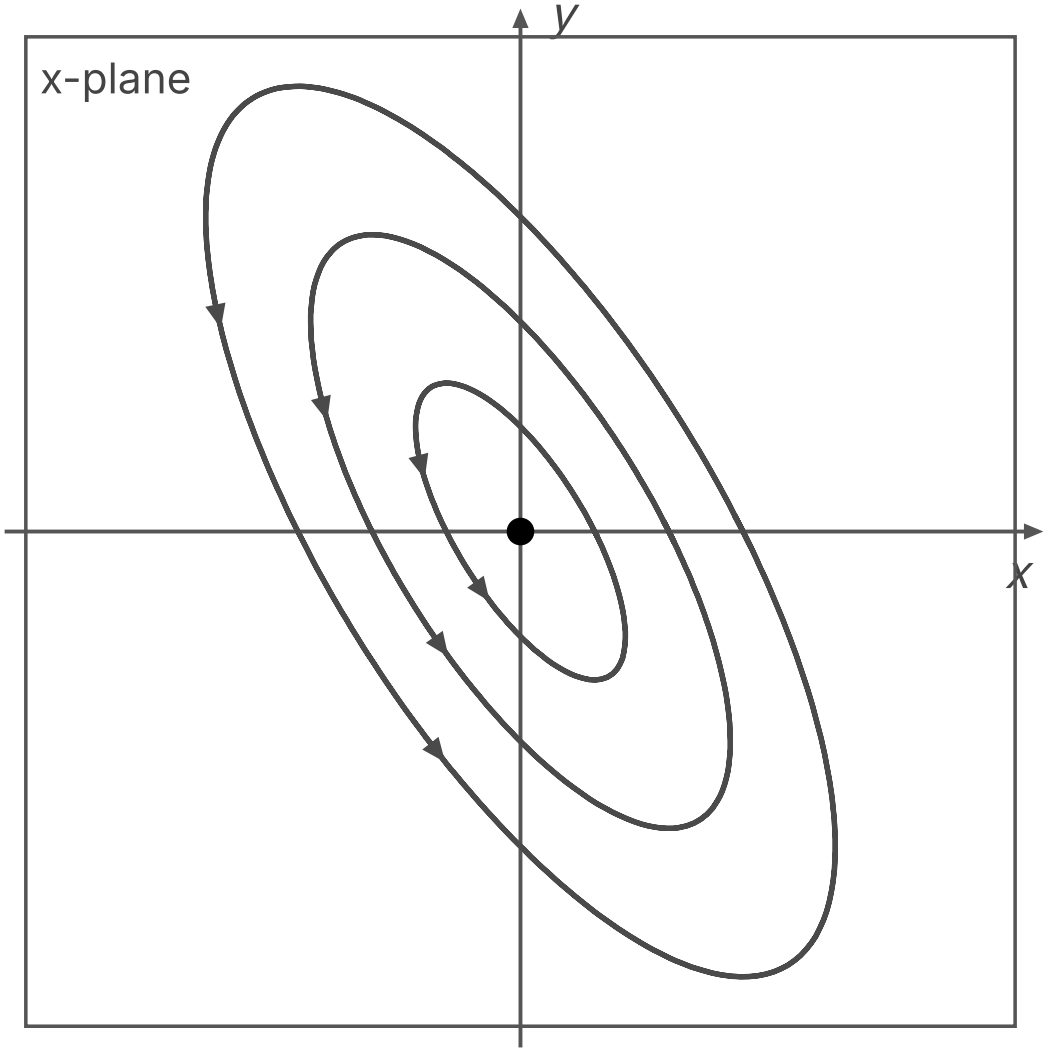
Phase portrait

$\rho = 0$ and $\omega = 3$: the origin is a **centre**. In the $\mathbf{z}$-plane $r = r_0 e^{0t}$ and $\theta = \theta_0 - 3t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and the orbits are circles of period $T = 2\pi/\omega$ (closed orbits).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}-3\\3\end{pmatrix}$, so $\dot{x} = -3$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: $\det Q = -5$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (centre): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (centre), counter-clockwise.

(iii) $\lambda_{\pm} = -2 \pm i$ with $\boldsymbol{\eta}_{\pm} = (1 \mp i,\ 3 \pm i),\ \dot{x} > 0$

Split the eigenvector of $\lambda_{+} = -2 + i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 1 - i \\ 3 + i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = -2, \omega = 1$ and

$$\mathbf{u} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 1 & -1 \\ 3 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} -2 & 1 \\ -1 & -2 \end{pmatrix}, \quad \det Q = 4, \quad Q^{-1} = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ -\frac{3}{4} & \frac{1}{4} \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 1\cdot(-2) + (-1)\cdot(-1) = -1\\
(QJ)_{12} &= 1\cdot 1 + (-1)\cdot(-2) = 3\\
(QJ)_{21} &= 3\cdot(-2) + 1\cdot(-1) = -7\\
(QJ)_{22} &= 3\cdot 1 + 1\cdot(-2) = 1
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -1 & 3 \\ -7 & 1 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-1)\cdot\frac{1}{4} + 3\cdot(-\frac{3}{4}) = -\frac{5}{2}\\
(A)_{12} &= (-1)\cdot\frac{1}{4} + 3\cdot\frac{1}{4} = \frac{1}{2}\\
(A)_{21} &= (-7)\cdot\frac{1}{4} + 1\cdot(-\frac{3}{4}) = -\frac{5}{2}\\
(A)_{22} &= (-7)\cdot\frac{1}{4} + 1\cdot\frac{1}{4} = -\frac{3}{2}
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} -\frac{5}{2} & \frac{1}{2} \\ -\frac{5}{2} & -\frac{3}{2} \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = \frac{1}{2}$, which is positive, in agreement with the statement ($\dot{x} > 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -1 \\ -7 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = (-2)\begin{pmatrix} 1 \\ 3 \end{pmatrix} - 1\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -7 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 1\begin{pmatrix} 1 \\ 3 \end{pmatrix} + (-2)\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 1 & -1 \\ 3 & 1 \end{pmatrix}
$$

$\det Q = (1)(1) - (-1)(3) = 4$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ -\frac{3}{4} & \frac{1}{4} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -1 & 3 \\ -7 & 1 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{1}{4}\cdot(-1) + \frac{1}{4}\cdot(-7) = -2 \\ (Q^{-1}AQ)_{12} &= \frac{1}{4}\cdot 3 + \frac{1}{4}\cdot 1 = 1 \\ (Q^{-1}AQ)_{21} &= (-\frac{3}{4})\cdot(-1) + \frac{1}{4}\cdot(-7) = -1 \\ (Q^{-1}AQ)_{22} &= (-\frac{3}{4})\cdot 3 + \frac{1}{4}\cdot 1 = -2 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -2 & 1 \\ -1 & -2 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = -2z_1 + z_2, \qquad \dot{z}_2 = -z_1 - 2z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0 \cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0 \cos\omega t + \sin\theta_0 \sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0 \cos\omega t - \cos\theta_0 \sin\omega t$:

$$z_1(t) = e^{-2t}\left(c_1 \cos 1t + c_2 \sin 1t\right), \qquad z_2(t) = e^{-2t}\left(c_2 \cos 1t - c_1 \sin 1t\right)$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1 \sin\omega t + c_2 \cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2 \sin\omega t - c_1 \cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

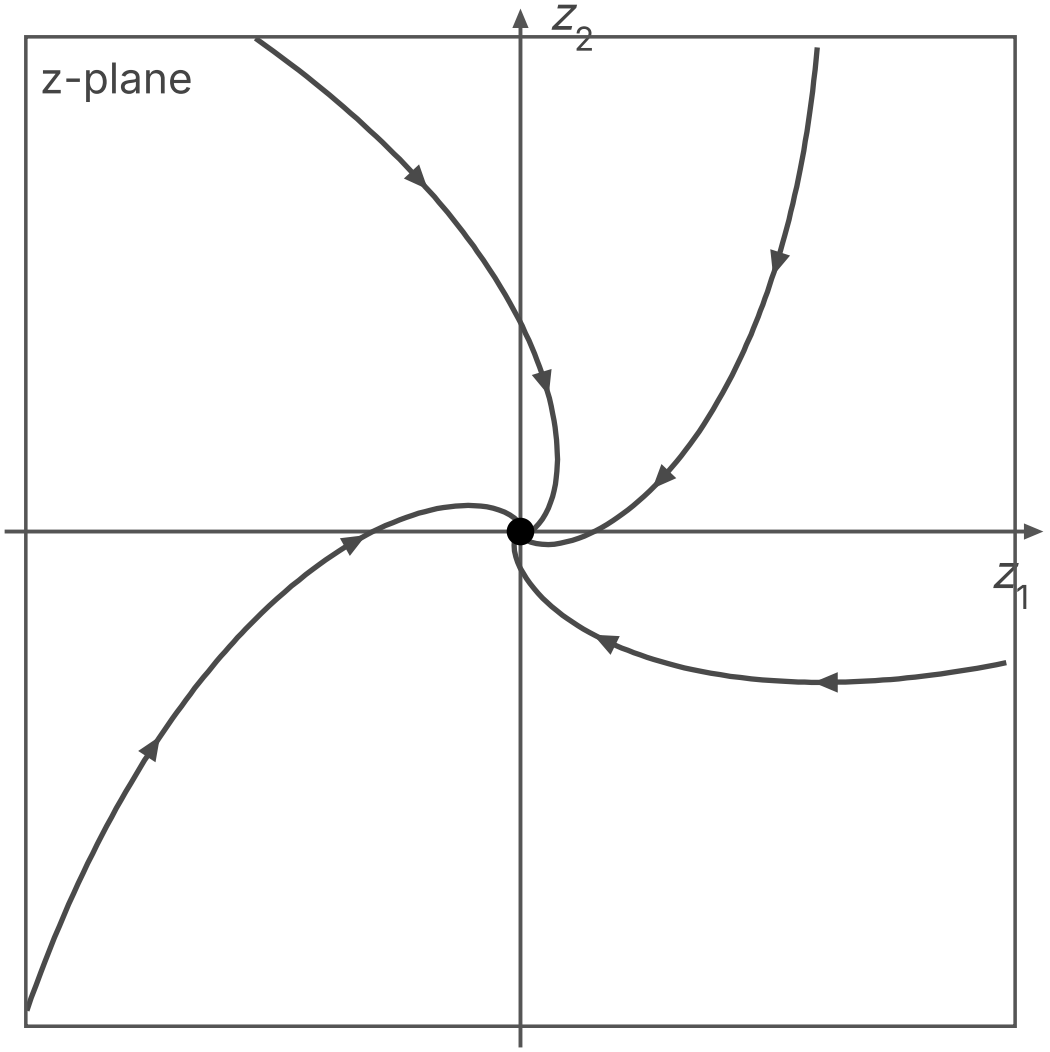
$$\mathbf{x}(t) = e^{-2t}\left[\left(c_1 \cos 1t + c_2 \sin 1t\right)\begin{pmatrix}1\\3\end{pmatrix} + \left(c_2 \cos 1t - c_1 \sin 1t\right)\begin{pmatrix}-1\\1\end{pmatrix}\right]$$

$$x(t) = c_1 e^{-2t}\sin t + c_1 e^{-2t}\cos t + c_2 e^{-2t}\sin t - c_2 e^{-2t}\cos t, \qquad y(t) = -c_1 e^{-2t}\sin t + 3c_1 e^{-2t}\cos t + 3c_2 e^{-2t}\sin t + c_2 e^{-2t}\cos t$$

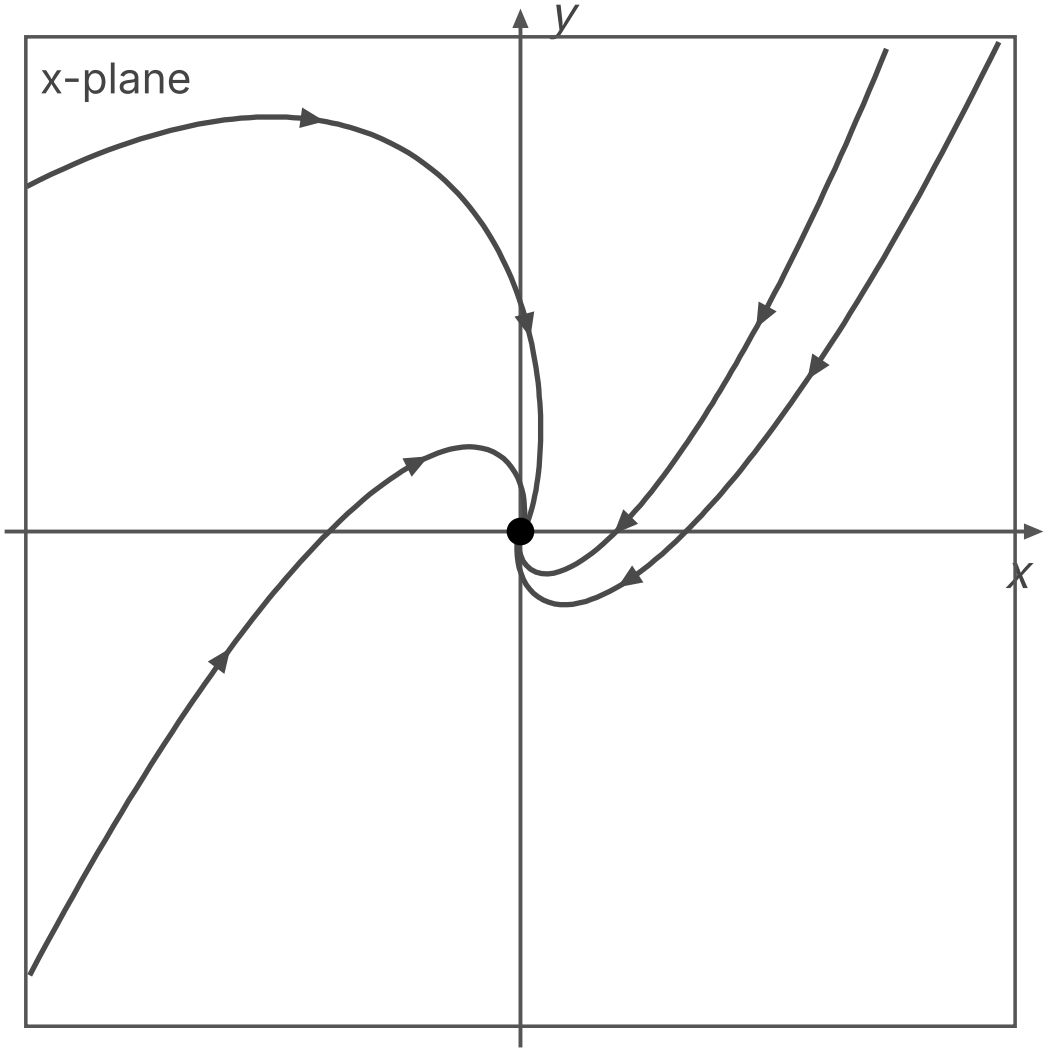
Phase portrait

$\rho = -2$ and $\omega = 1$: the origin is a **stable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{-2t}$ and $\theta = \theta_0 - 1t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{-4\pi}$ (shrinking).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}\frac{1}{2}\\ -\frac{3}{2}\end{pmatrix}$, so $\dot{x} = \frac{1}{2}$ is > 0: to the right at the top of the plane: the rotation is **clockwise**. Cross-check: $\det Q = 4$ is > 0, so Q preserves orientation and the clockwise motion of the $\mathbf{z}$-plane becomes clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (stable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (stable spiral), clockwise.

(iv) $\lambda_{\pm} = -1 \pm i$ with $\boldsymbol{\eta}_{\pm} = (1,\ \pm i),\ \dot{x} > 0$

Split the eigenvector of $\lambda_{+} = -1 + i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 1 \\ i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = -1, \omega = 1$ and

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}, \quad \det Q = 1, \quad Q^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 1\cdot(-1) + 0\cdot(-1) = -1\\
(QJ)_{12} &= 1\cdot 1 + 0\cdot(-1) = 1\\
(QJ)_{21} &= 0\cdot(-1) + 1\cdot(-1) = -1\\
(QJ)_{22} &= 0\cdot 1 + 1\cdot(-1) = -1
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-1)\cdot 1 + 1\cdot 0 = -1\\
(A)_{12} &= (-1)\cdot 0 + 1\cdot 1 = 1\\
(A)_{21} &= (-1)\cdot 1 + (-1)\cdot 0 = -1\\
(A)_{22} &= (-1)\cdot 0 + (-1)\cdot 1 = -1
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = 1$, which is positive, in agreement with the statement ($\dot{x} > 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = (-1)\begin{pmatrix} 1 \\ 0 \end{pmatrix} - 1\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} + (-1)\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}, \mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}
$$

$\det Q = (1)(1) - (0)(0) = 1$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{1}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= 1 \cdot (-1) + 0 \cdot (-1) = -1 \\ (Q^{-1}AQ)_{12} &= 1 \cdot 1 + 0 \cdot (-1) = 1 \\ (Q^{-1}AQ)_{21} &= 0 \cdot (-1) + 1 \cdot (-1) = -1 \\ (Q^{-1}AQ)_{22} &= 0 \cdot 1 + 1 \cdot (-1) = -1 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = -z_1 + z_2, \qquad \dot{z}_2 = -z_1 - z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot r - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0 \cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0 \cos\omega t + \sin\theta_0 \sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0 \cos\omega t - \cos\theta_0 \sin\omega t$:

$$z_1(t) = e^{-1t}\left(c_1 \cos 1t + c_2 \sin 1t\right), \qquad z_2(t) = e^{-1t}\left(c_2 \cos 1t - c_1 \sin 1t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1 \sin\omega t + c_2 \cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2 \sin\omega t - c_1 \cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

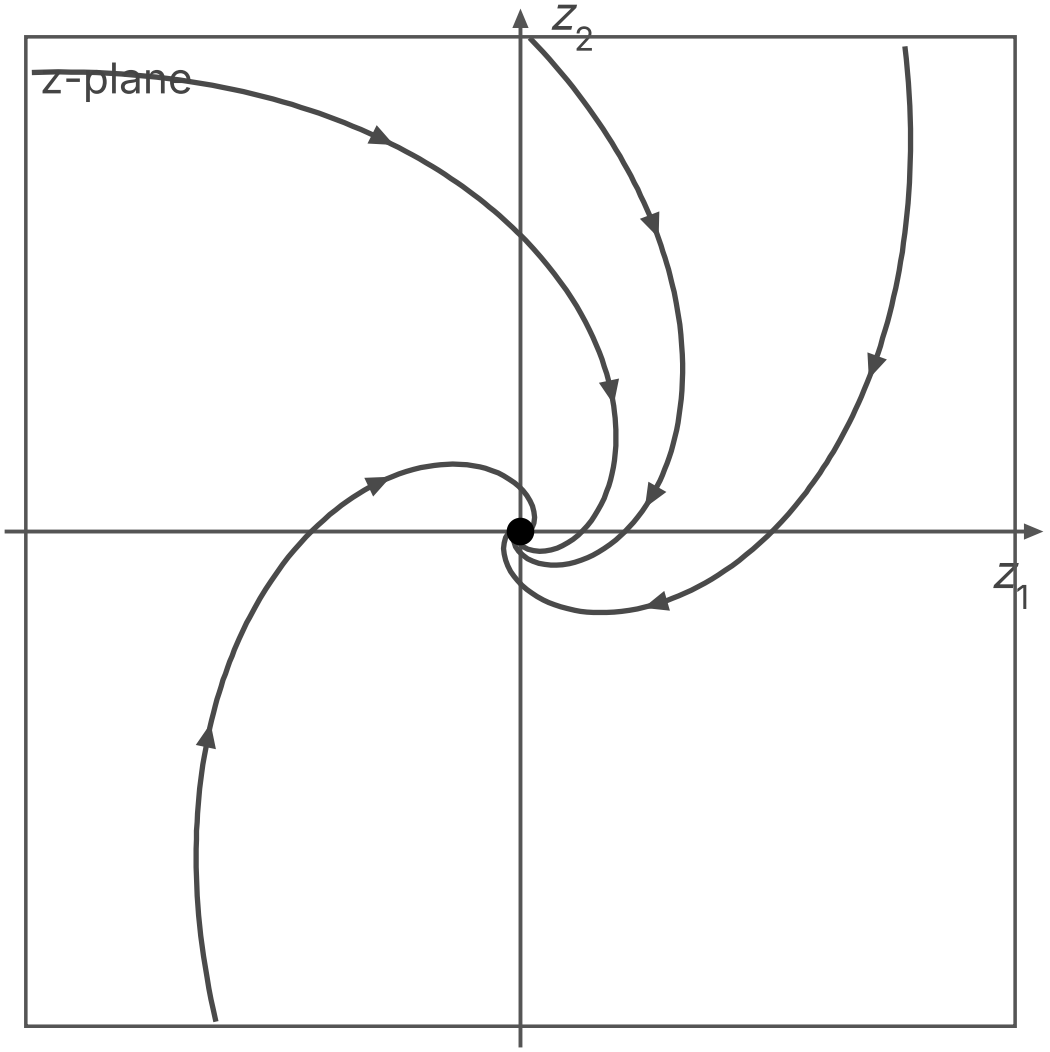
$$\mathbf{x}(t) = e^{-1t}\left[\left(c_1 \cos 1t + c_2 \sin 1t\right)\begin{pmatrix}1\\0\end{pmatrix} + \left(c_2 \cos 1t - c_1 \sin 1t\right)\begin{pmatrix}0\\1\end{pmatrix}\right]$$

$$x(t) = c_1 e^{-t}\cos t + c_2 e^{-t}\sin t, \qquad y(t) = -c_1 e^{-t}\sin t + c_2 e^{-t}\cos t$$

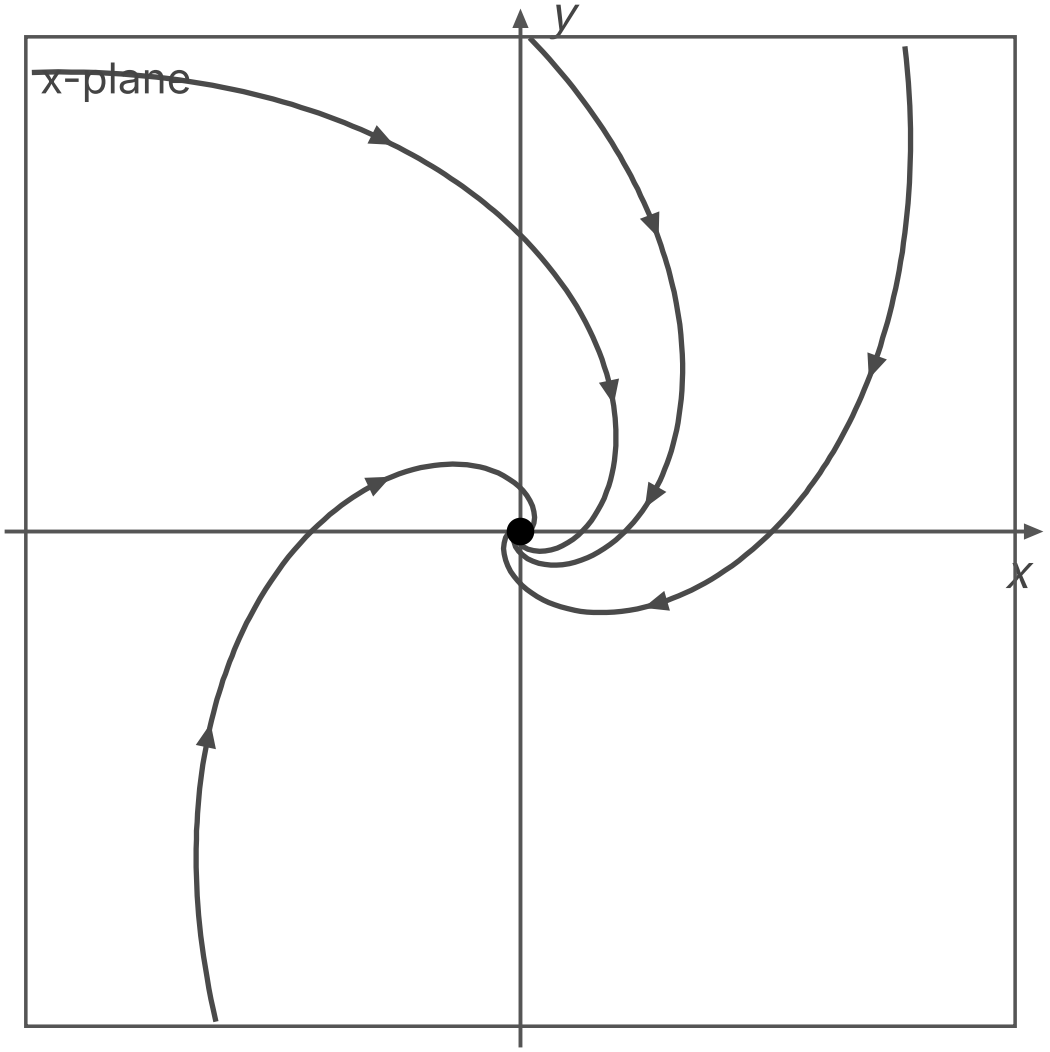
Phase portrait

$\rho = -1$ and $\omega = 1$: the origin is a **stable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{-1t}$ and $\theta = \theta_0 - 1t$: the motion is always **clockwise** ($\dot\theta = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{-2\pi}$ (shrinking).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}1\\-1\end{pmatrix}$, so $\dot x = 1$ is > 0: to the right at the top of the plane: the rotation is **clockwise**. Cross-check: $\det Q = 1$ is > 0, so Q preserves orientation and the clockwise motion of the $\mathbf{z}$-plane becomes clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (stable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (stable spiral), clockwise.

(v) $\lambda_{\pm} = 2 \pm 2i$ with $\boldsymbol{\eta}_{\pm} = (\pm 3i,\ 5 \mp 4i)$, $\dot{x} < 0$

Split the eigenvector of $\lambda_{+} = 2 + 2i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 3i \\ 5 - 4i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = 2, \omega = 2$ and

$$\mathbf{u} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 0 & 3 \\ 5 & -4 \end{pmatrix}, \quad J = \begin{pmatrix} 2 & 2 \\ -2 & 2 \end{pmatrix}, \quad \det Q = -15, \quad Q^{-1} = \begin{pmatrix} \frac{4}{15} & \frac{1}{5} \\ \frac{1}{3} & 0 \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 0\cdot 2 + 3\cdot(-2) = -6\\
(QJ)_{12} &= 0\cdot 2 + 3\cdot 2 = 6\\
(QJ)_{21} &= 5\cdot 2 + (-4)\cdot(-2) = 18\\
(QJ)_{22} &= 5\cdot 2 + (-4)\cdot 2 = 2
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -6 & 6 \\ 18 & 2 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-6)\cdot\frac{4}{15} + 6\cdot\frac{1}{3} = \frac{2}{5}\\
(A)_{12} &= (-6)\cdot\frac{1}{5} + 6\cdot 0 = -\frac{6}{5}\\
(A)_{21} &= 18\cdot\frac{4}{15} + 2\cdot\frac{1}{3} = \frac{82}{15}\\
(A)_{22} &= 18\cdot\frac{1}{5} + 2\cdot 0 = \frac{18}{5}
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} \frac{2}{5} & -\frac{6}{5} \\ \frac{82}{15} & \frac{18}{5} \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = -\frac{6}{5}$, which is negative, in agreement with the statement ($\dot{x} < 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -6 \\ 18 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 2\begin{pmatrix} 0 \\ 5 \end{pmatrix} - 2\begin{pmatrix} 3 \\ -4 \end{pmatrix} = \begin{pmatrix} -6 \\ 18 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 2\begin{pmatrix} 0 \\ 5 \end{pmatrix} + 2\begin{pmatrix} 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 0 & 3 \\ 5 & -4 \end{pmatrix}
$$

$\det Q = (0)(-4) - (3)(5) = -15$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-15}\begin{pmatrix} -4 & -3 \\ -5 & 0 \end{pmatrix} = \begin{pmatrix} \frac{4}{15} & \frac{1}{5} \\ \frac{1}{3} & 0 \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -6 & 6 \\ 18 & 2 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{4}{15}\cdot(-6) + \frac{1}{5}\cdot 18 = 2 \\ (Q^{-1}AQ)_{12} &= \frac{4}{15}\cdot 6 + \frac{1}{5}\cdot 2 = 2 \\ (Q^{-1}AQ)_{21} &= \frac{1}{3}\cdot(-6) + 0\cdot 18 = -2 \\ (Q^{-1}AQ)_{22} &= \frac{1}{3}\cdot 6 + 0\cdot 2 = 2 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 2 & 2 \\ -2 & 2 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = 2z_1 + 2z_2, \qquad \dot{z}_2 = -2z_1 + 2z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot z_2 - z_2\dot z_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot\theta = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot r - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{2t}\left(c_1\cos 2t + c_2 \sin 2t\right), \qquad z_2(t) = e^{2t}\left(c_2\cos 2t - c_1\sin 2t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

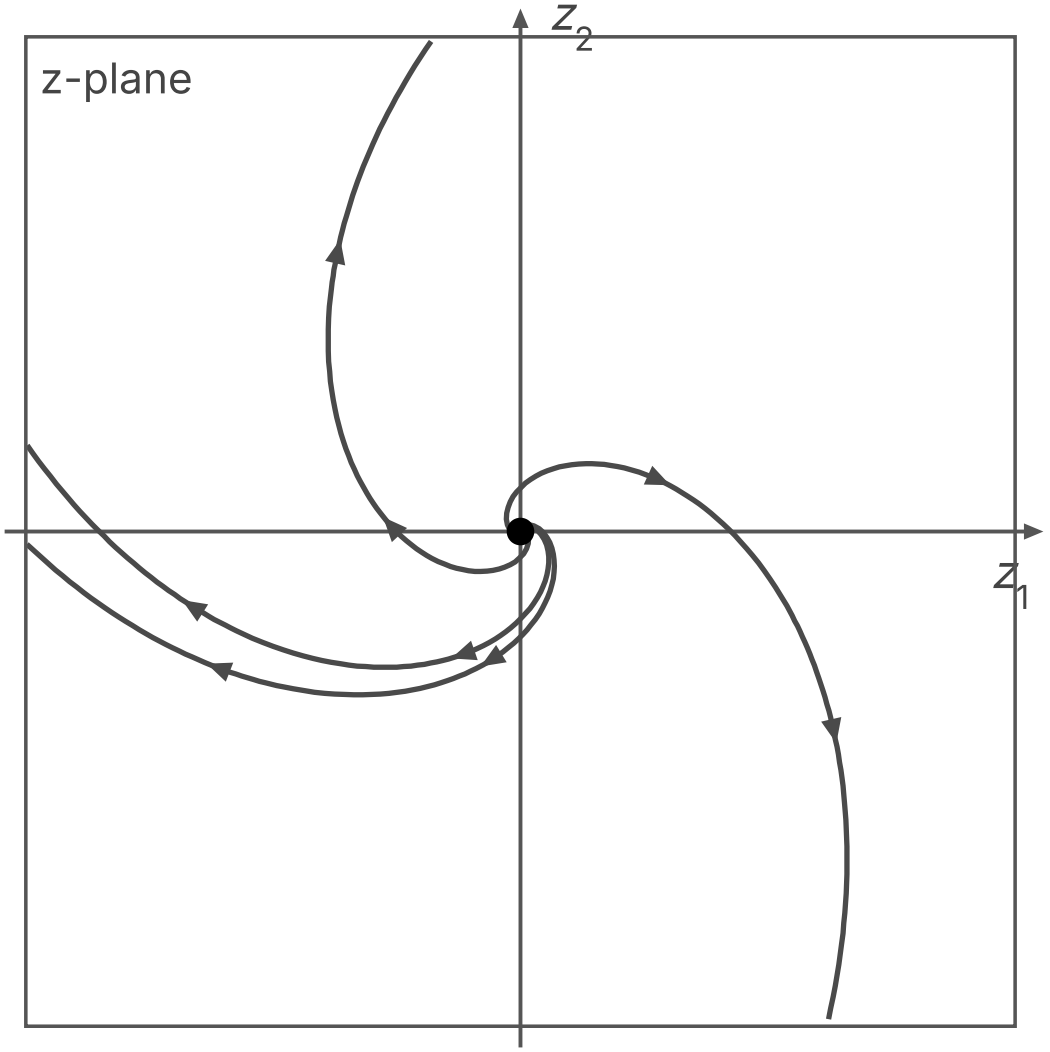
$$\mathbf{x}(t) = e^{2t}\left[(c_1\cos 2t + c_2\sin 2t)\begin{pmatrix}0\\5\end{pmatrix} + (c_2\cos 2t - c_1\sin 2t)\begin{pmatrix}3\\-4\end{pmatrix}\right]$$

$$x(t) = -3c_1 e^{2t}\sin 2t + 3c_2 e^{2t}\cos 2t, \qquad y(t) = 4c_1 e^{2t}\sin 2t + 5c_1 e^{2t}\cos 2t + 5c_2 e^{2t}\sin 2t - 4c_2 e^{2t}\cos 2t$$

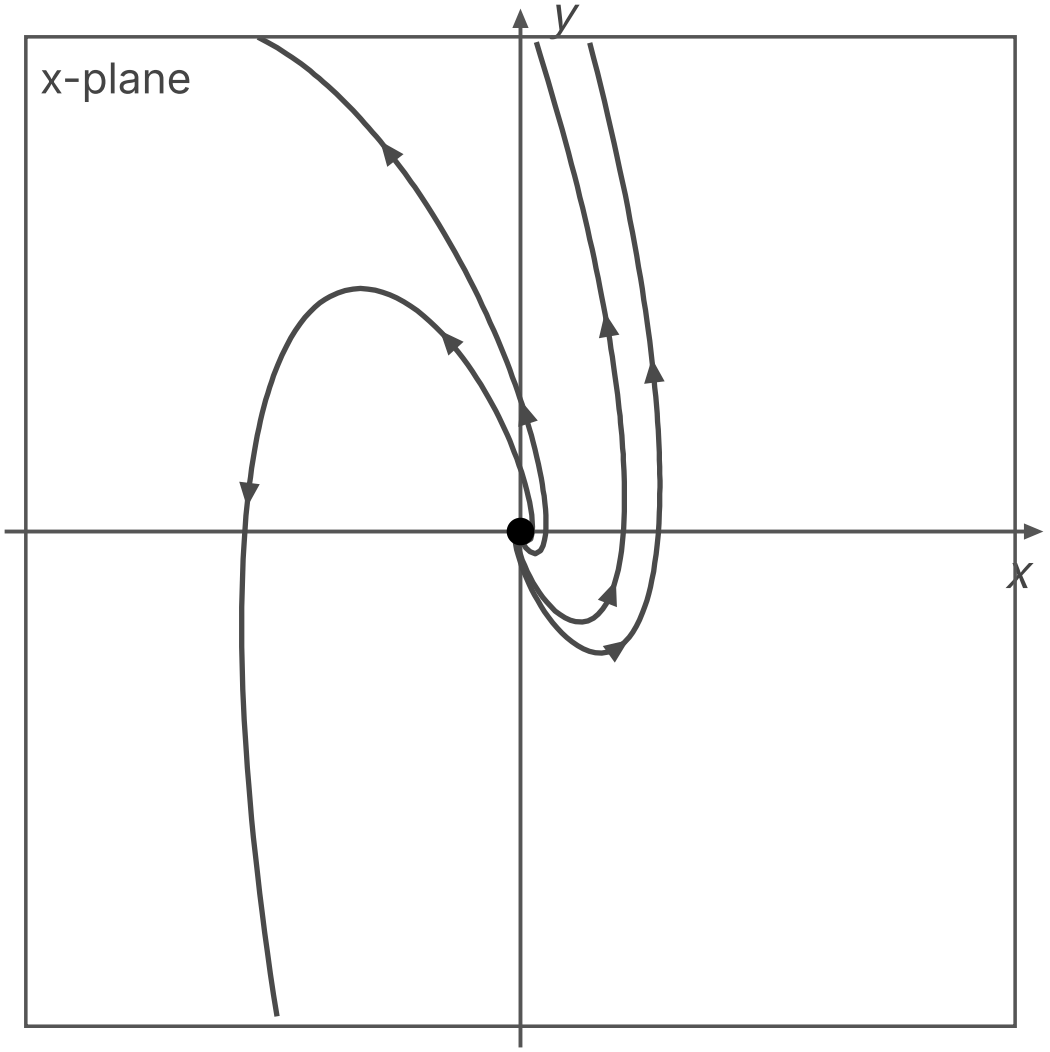
Phase portrait

$\rho = 2$ and $\omega = 2$: the origin is a **unstable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{2t}$ and $\theta = \theta_0 - 2t$: the motion is always **clockwise** ($\dot\theta = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{2\pi}$ (growing).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}-\frac{6}{5}\\ \frac{18}{5}\end{pmatrix}$, so $\dot x = -\frac{6}{5}$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: $\det Q = -15$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (unstable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (unstable spiral), counter-clockwise.

(vi) $\lambda_{\pm} = 5 \pm 3i$ with $\boldsymbol{\eta}_{\pm} = (2 \pm 5i,\ \mp i),\ \dot{x} < 0$

Split the eigenvector of $\lambda_{+} = 5 + 3i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 2 + 5i \\ -i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = 5, \omega = 3$ and

$$\mathbf{u} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 2 & 5 \\ 0 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 5 & 3 \\ -3 & 5 \end{pmatrix}, \quad \det Q = -2, \quad Q^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{5}{2} \\ 0 & -1 \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 2\cdot 5 + 5\cdot(-3) = -5\\
(QJ)_{12} &= 2\cdot 3 + 5\cdot 5 = 31\\
(QJ)_{21} &= 0\cdot 5 + (-1)\cdot(-3) = 3\\
(QJ)_{22} &= 0\cdot 3 + (-1)\cdot 5 = -5
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -5 & 31 \\ 3 & -5 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-5)\cdot\frac{1}{2} + 31\cdot 0 = -\frac{5}{2}\\
(A)_{12} &= (-5)\cdot\frac{5}{2} + 31\cdot(-1) = -\frac{87}{2}\\
(A)_{21} &= 3\cdot\frac{1}{2} + (-5)\cdot 0 = \frac{3}{2}\\
(A)_{22} &= 3\cdot\frac{5}{2} + (-5)\cdot(-1) = \frac{25}{2}
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} -\frac{5}{2} & -\frac{87}{2} \\ \frac{3}{2} & \frac{25}{2} \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = -\frac{87}{2}$, which is negative, in agreement with the statement ($\dot{x} < 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -5 \\ 3 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 5\begin{pmatrix} 2 \\ 0 \end{pmatrix} - 3\begin{pmatrix} 5 \\ -1 \end{pmatrix} = \begin{pmatrix} -5 \\ 3 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 31 \\ -5 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 3\begin{pmatrix} 2 \\ 0 \end{pmatrix} + 5\begin{pmatrix} 5 \\ -1 \end{pmatrix} = \begin{pmatrix} 31 \\ -5 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 2 & 5 \\ 0 & -1 \end{pmatrix}
$$

$\det Q = (2)(-1) - (5)(0) = -2$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{-2}\begin{pmatrix} -1 & -5 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{5}{2} \\ 0 & -1 \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -5 & 31 \\ 3 & -5 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{1}{2}\cdot(-5) + \frac{5}{2}\cdot 3 = 5 \\ (Q^{-1}AQ)_{12} &= \frac{1}{2}\cdot 31 + \frac{5}{2}\cdot(-5) = 3 \\ (Q^{-1}AQ)_{21} &= 0\cdot(-5) + (-1)\cdot 3 = -3 \\ (Q^{-1}AQ)_{22} &= 0\cdot 31 + (-1)\cdot(-5) = 5 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 5 & 3 \\ -3 & 5 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = 5z_1 + 3z_2, \qquad \dot{z}_2 = -3z_1 + 5z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0 \cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0 \cos\omega t + \sin\theta_0 \sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0 \cos\omega t - \cos\theta_0 \sin\omega t$:

$$z_1(t) = e^{5t}\left(c_1 \cos 3t + c_2 \sin 3t\right), \qquad z_2(t) = e^{5t}\left(c_2 \cos 3t - c_1 \sin 3t\right)$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1 \sin\omega t + c_2 \cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2 \sin\omega t - c_1 \cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

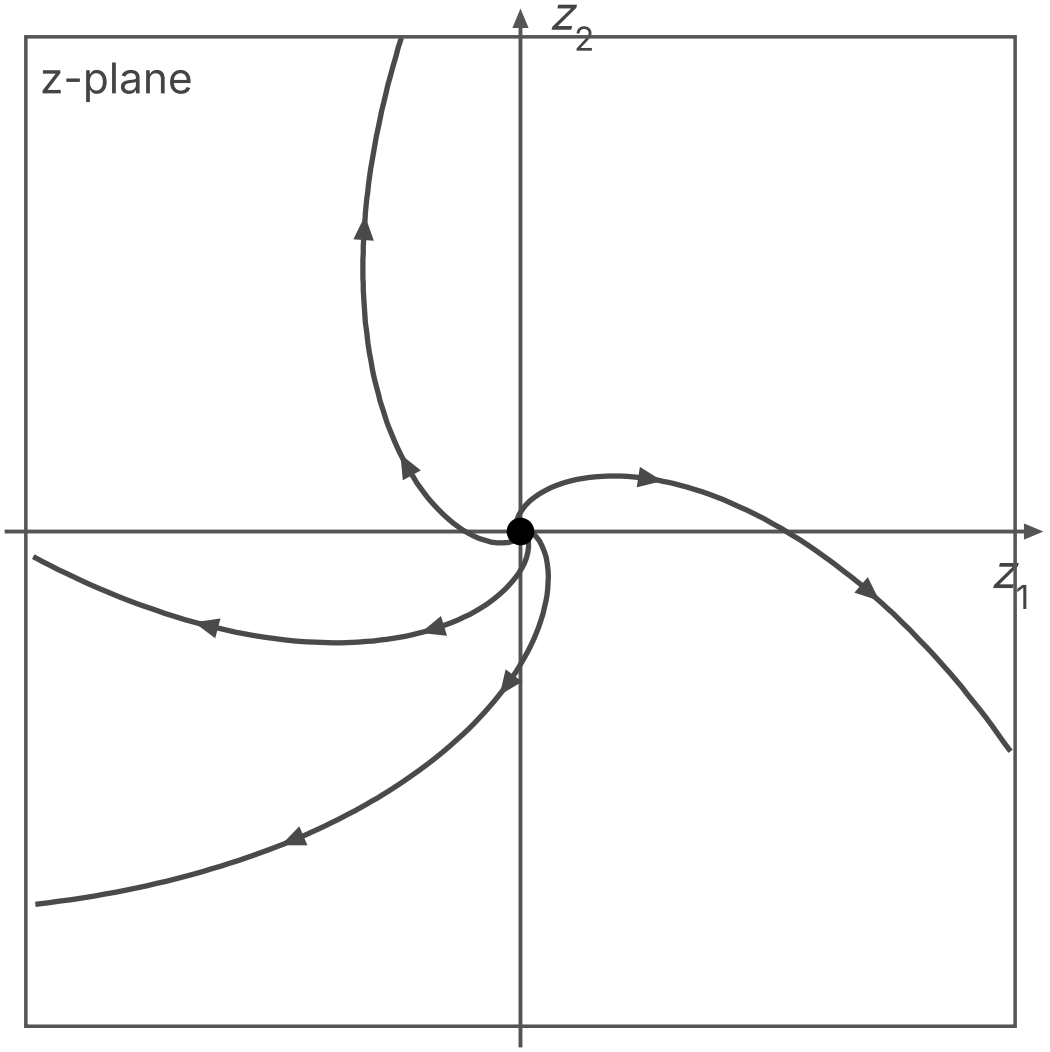
$$\mathbf{x}(t) = e^{5t}\left[\left(c_1 \cos 3t + c_2 \sin 3t\right)\begin{pmatrix}2\\0\end{pmatrix} + \left(c_2 \cos 3t - c_1 \sin 3t\right)\begin{pmatrix}5\\-1\end{pmatrix}\right]$$

$$x(t) = -5c_1 e^{5t}\sin 3t + 2c_1 e^{5t}\cos 3t + 2c_2 e^{5t}\sin 3t + 5c_2 e^{5t}\cos 3t, \qquad y(t) = c_1 e^{5t}\sin 3t - c_2 e^{5t}\cos 3t$$

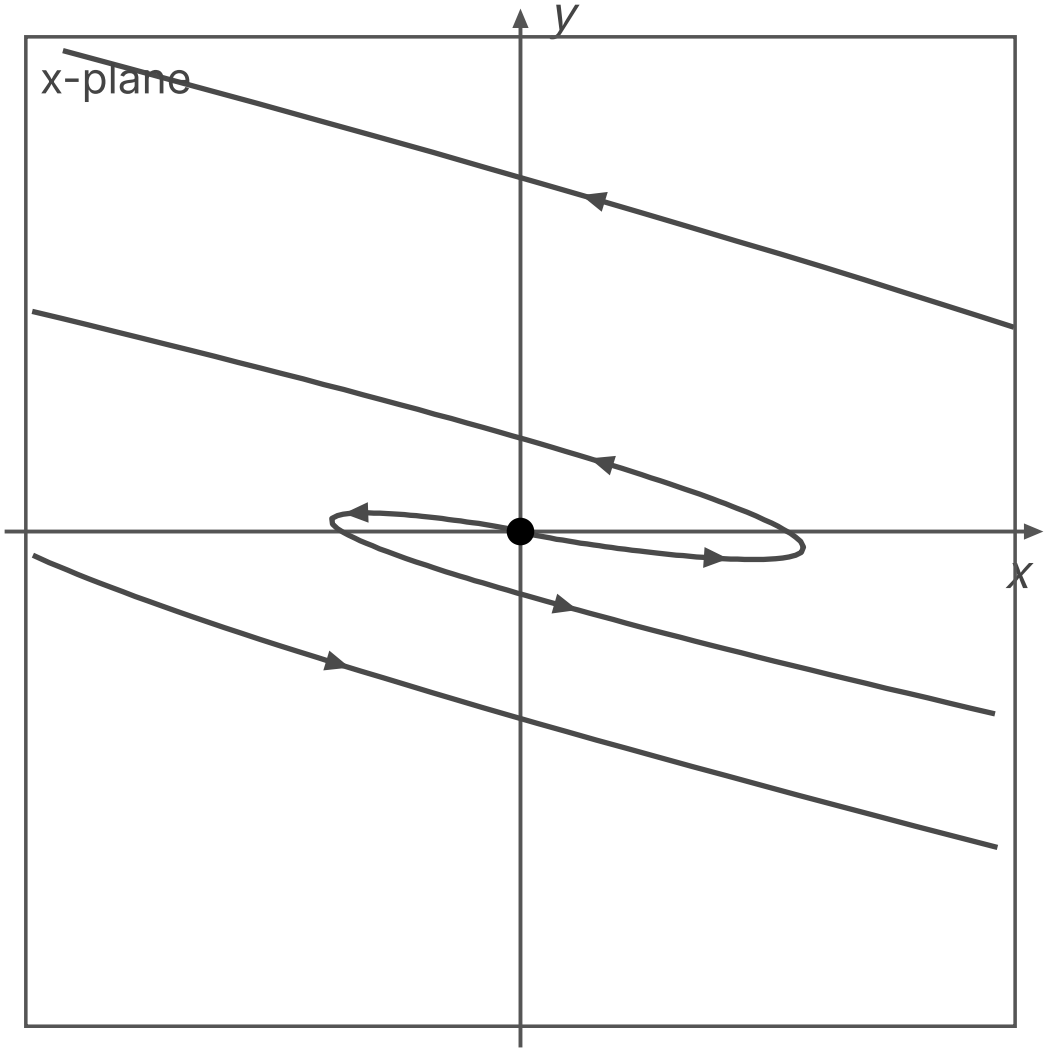
Phase portrait

$\rho = 5$ and $\omega = 3$: the origin is a **unstable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{5t}$ and $\theta = \theta_0 - 3t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{\frac{10\pi}{3}}$ (growing).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}-\frac{87}{2}\\ \frac{25}{2}\end{pmatrix}$, so $\dot{x} = -\frac{87}{2}$ is < 0: to the left at the top of the plane: the rotation is **counter-clockwise**. Cross-check: $\det Q = -2$ is < 0, so Q reverses orientation and the clockwise motion of the $\mathbf{z}$-plane becomes counter-clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (unstable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (unstable spiral), counter-clockwise.

(vii) $\lambda_{\pm} = \pm 7i$ with $\boldsymbol{\eta}_{\pm} = (1 \pm i,\ -1 \pm 2i)$, $\dot{x} > 0$

Split the eigenvector of $\lambda_{+} = 7i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} 1+i \\ -1+2i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = 0, \omega = 7$ and

$$\mathbf{u} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 7 \\ -7 & 0 \end{pmatrix}, \quad \det Q = 3, \quad Q^{-1} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= 1\cdot 0 + 1\cdot(-7) = -7\\
(QJ)_{12} &= 1\cdot 7 + 1\cdot 0 = 7\\
(QJ)_{21} &= (-1)\cdot 0 + 2\cdot(-7) = -14\\
(QJ)_{22} &= (-1)\cdot 7 + 2\cdot 0 = -7
\end{aligned}
$$

$$
QJ = \begin{pmatrix} -7 & 7 \\ -14 & -7 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= (-7)\cdot\frac{2}{3} + 7\cdot\frac{1}{3} = -\frac{7}{3}\\
(A)_{12} &= (-7)\cdot(-\frac{1}{3}) + 7\cdot\frac{1}{3} = \frac{14}{3}\\
(A)_{21} &= (-14)\cdot\frac{2}{3} + (-7)\cdot\frac{1}{3} = -\frac{35}{3}\\
(A)_{22} &= (-14)\cdot(-\frac{1}{3}) + (-7)\cdot\frac{1}{3} = \frac{7}{3}
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} -\frac{7}{3} & \frac{14}{3} \\ -\frac{35}{3} & \frac{7}{3} \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = \frac{14}{3}$, which is positive, in agreement with the statement ($\dot{x} > 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} -7 \\ -14 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = 0\begin{pmatrix} 1 \\ -1 \end{pmatrix} - 7\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -7 \\ -14 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} 7 \\ -7 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 7\begin{pmatrix} 1 \\ -1 \end{pmatrix} + 0\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 7 \\ -7 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\mathbf{u} \mid \mathbf{w}] = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}
$$

$\det Q = (1)(2) - (1)(-1) = 3$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} -7 & 7 \\ -14 & -7 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= \frac{2}{3}\cdot(-7) + (-\frac{1}{3})\cdot(-14) = 0 \\ (Q^{-1}AQ)_{12} &= \frac{2}{3}\cdot 7 + (-\frac{1}{3})\cdot(-7) = 7 \\ (Q^{-1}AQ)_{21} &= \frac{1}{3}\cdot(-7) + \frac{1}{3}\cdot(-14) = -7 \\ (Q^{-1}AQ)_{22} &= \frac{1}{3}\cdot 7 + \frac{1}{3}\cdot(-7) = 0 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} 0 & 7 \\ -7 & 0 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = 7z_2, \qquad \dot{z}_2 = -7z_1$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot{z}_2 - z_2\dot{z}_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot{\theta} = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot{r} - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0 \cos\theta_0 = z_1(0)$, $c_2 = r_0 \sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0 \cos\omega t + \sin\theta_0 \sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0 \cos\omega t - \cos\theta_0 \sin\omega t$:

$$z_1(t) = e^{0t}\left(c_1 \cos 7t + c_2 \sin 7t\right), \qquad z_2(t) = e^{0t}\left(c_2 \cos 7t - c_1 \sin 7t\right)$$

Check by substitution (product rule and chain rule): $\dot{z}_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1 \sin\omega t + c_2 \cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot{z}_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2 \sin\omega t - c_1 \cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

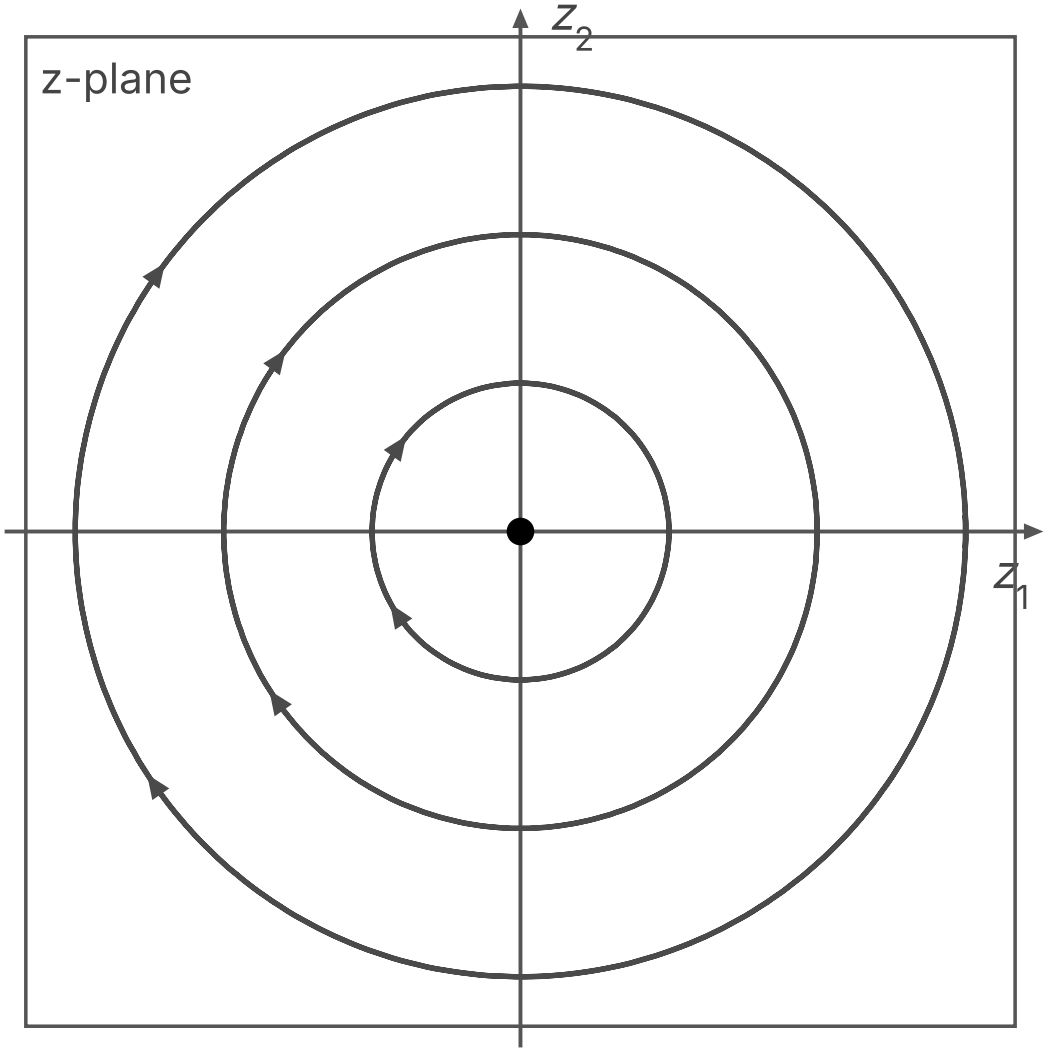
$$\mathbf{x}(t) = e^{0t}\left[\left(c_1 \cos 7t + c_2 \sin 7t\right)\begin{pmatrix}1\\-1\end{pmatrix} + \left(c_2 \cos 7t - c_1 \sin 7t\right)\begin{pmatrix}1\\2\end{pmatrix}\right]$$

$$x(t) = -c_1 \sin 7t + c_1 \cos 7t + c_2 \sin 7t + c_2 \cos 7t, \qquad y(t) = -2c_1 \sin 7t - c_1 \cos 7t - c_2 \sin 7t + 2c_2 \cos 7t$$

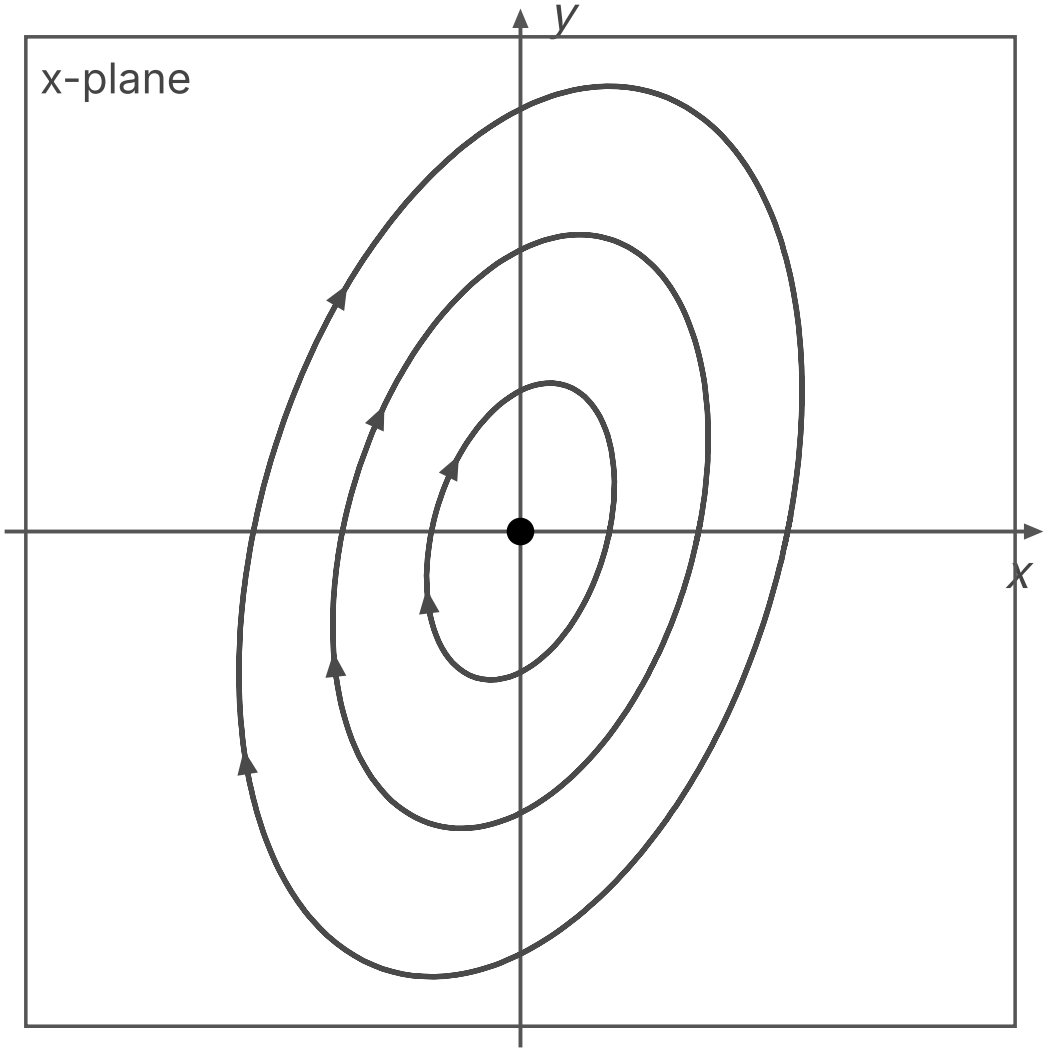
Phase portrait

$\rho = 0$ and $\omega = 7$: the origin is a **centre**. In the $\mathbf{z}$-plane $r = r_0 e^{0t}$ and $\theta = \theta_0 - 7t$: the motion is always **clockwise** ($\dot{\theta} = -\omega < 0$), and the orbits are circles of period $T = 2\pi/\omega$ (closed orbits).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}\frac{14}{3}\\ \frac{7}{3}\end{pmatrix}$, so $\dot{x} = \frac{14}{3}$ is > 0: to the right at the top of the plane: the rotation is **clockwise**. Cross-check: $\det Q = 3$ is > 0, so Q preserves orientation and the clockwise motion of the $\mathbf{z}$-plane becomes clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (centre): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (centre), clockwise.

(viii) $\lambda_{\pm} = -13 \pm 17i$ with $\boldsymbol{\eta}_{\pm} = (\pm 6i - 8,\ 4 \mp 5i)$, $\dot{x} > 0$

Split the eigenvector of $\lambda_{+} = -13 + 17i$ into real and imaginary parts, $\boldsymbol{\eta}_{+} = \begin{pmatrix} -8 + 6i \\ 4 - 5i \end{pmatrix} = \mathbf{u} + i\mathbf{w}$, so $\rho = -13, \omega = 17$ and

$$\mathbf{u} = \begin{pmatrix} -8 \\ 4 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 6 \\ -5 \end{pmatrix}$$

The matrix is not given; it is rebuilt from $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$, $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$ (Topic 4 Eq. 7.4), which say $AQ = QJ$ with $Q = [\mathbf{u} \mid \mathbf{w}]$, so $A = QJQ^{-1}$:

$$Q = \begin{pmatrix} -8 & 6 \\ 4 & -5 \end{pmatrix}, \quad J = \begin{pmatrix} -13 & 17 \\ -17 & -13 \end{pmatrix}, \quad \det Q = 16, \quad Q^{-1} = \begin{pmatrix} -\frac{5}{16} & -\frac{3}{8} \\ -\frac{1}{4} & -\frac{1}{2} \end{pmatrix}$$

$$
\begin{aligned}
(QJ)_{11} &= (-8)\cdot(-13) + 6\cdot(-17) = 2\\
(QJ)_{12} &= (-8)\cdot 17 + 6\cdot(-13) = -214\\
(QJ)_{21} &= 4\cdot(-13) + (-5)\cdot(-17) = 33\\
(QJ)_{22} &= 4\cdot 17 + (-5)\cdot(-13) = 133
\end{aligned}
$$

$$
QJ = \begin{pmatrix} 2 & -214 \\ 33 & 133 \end{pmatrix}
$$

$$
\begin{aligned}
(A)_{11} &= 2\cdot(-\frac{5}{16}) + (-214)\cdot(-\frac{1}{4}) = \frac{423}{8}\\
(A)_{12} &= 2\cdot(-\frac{3}{8}) + (-214)\cdot(-\frac{1}{2}) = \frac{425}{4}\\
(A)_{21} &= 33\cdot(-\frac{5}{16}) + 133\cdot(-\frac{1}{4}) = -\frac{697}{16}\\
(A)_{22} &= 33\cdot(-\frac{3}{8}) + 133\cdot(-\frac{1}{2}) = -\frac{631}{8}
\end{aligned}
$$

$$
A = QJQ^{-1} = \begin{pmatrix} \frac{423}{8} & \frac{425}{4} \\ -\frac{697}{16} & -\frac{631}{8} \end{pmatrix}
$$

Check of the given sign: $\dot{x}$ at $(0,1)$ equals $a_{12} = \frac{425}{4}$, which is positive, in agreement with the statement ($\dot{x} > 0$).

How A acts on $\mathbf{u}$ and $\mathbf{w}$

Separating the real and imaginary parts of $A\mathbf{v} = \lambda\mathbf{v}$ gives (Topic 4, Eq. 7.4) $A\mathbf{u} = \rho\mathbf{u} - \omega\mathbf{w}$ and $A\mathbf{w} = \omega\mathbf{u} + \rho\mathbf{w}$. Verification with the numbers:

$$
A\mathbf{u} = \begin{pmatrix} 2 \\ 33 \end{pmatrix}, \qquad \rho\mathbf{u} - \omega\mathbf{w} = (-13)\begin{pmatrix} -8 \\ 4 \end{pmatrix} - 17\begin{pmatrix} 6 \\ -5 \end{pmatrix} = \begin{pmatrix} 2 \\ 33 \end{pmatrix}
$$

$$
A\mathbf{w} = \begin{pmatrix} -214 \\ 133 \end{pmatrix}, \qquad \omega\mathbf{u} + \rho\mathbf{w} = 17\begin{pmatrix} -8 \\ 4 \end{pmatrix} + (-13)\begin{pmatrix} 6 \\ -5 \end{pmatrix} = \begin{pmatrix} -214 \\ 133 \end{pmatrix}
$$

Change of coordinates

$\mathbf{u}$ and $\mathbf{w}$ are linearly independent (they are not parallel when $\omega \neq 0$; Topic 4, Lemma of Sec. 7.3). With the columns in the order $\mathbf{u}$, $\mathbf{w}$,

$$
Q = [\,\mathbf{u} \mid \mathbf{w}\,] = \begin{pmatrix} -8 & 6 \\ 4 & -5 \end{pmatrix}
$$

$\det Q = (-8)(-5) - (6)(4) = 16$, which is not zero, so Q is invertible. For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ (Robinson, Appendix B):

$$Q^{-1} = \frac{1}{16}\begin{pmatrix} -5 & -6 \\ -4 & -8 \end{pmatrix} = \begin{pmatrix} -\frac{5}{16} & -\frac{3}{8} \\ -\frac{1}{4} & -\frac{1}{2} \end{pmatrix}$$

Compute $AQ = [A\mathbf{u} \mid A\mathbf{w}]$ and multiply by Q^{-1} on the left, entry by entry:

$$AQ = \begin{pmatrix} 2 & -214 \\ 33 & 133 \end{pmatrix}$$

$$\begin{aligned} (Q^{-1}AQ)_{11} &= (-\frac{5}{16}) \cdot 2 + (-\frac{3}{8}) \cdot 33 = -13 \\ (Q^{-1}AQ)_{12} &= (-\frac{5}{16}) \cdot (-214) + (-\frac{3}{8}) \cdot 133 = 17 \\ (Q^{-1}AQ)_{21} &= (-\frac{1}{4}) \cdot 2 + (-\frac{1}{2}) \cdot 33 = -17 \\ (Q^{-1}AQ)_{22} &= (-\frac{1}{4}) \cdot (-214) + (-\frac{1}{2}) \cdot 133 = -13 \end{aligned}$$

$$Q^{-1}AQ = \begin{pmatrix} -13 & 17 \\ -17 & -13 \end{pmatrix} = J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$$

Solution in the canonical plane

$$\dot{z}_1 = -13z_1 + 17z_2, \qquad \dot{z}_2 = -17z_1 - 13z_2$$

The two equations are coupled, but they say "scale and rotate", which polar coordinates express directly: $z_1 = r\cos\theta$, $z_2 = r\sin\theta$ (Topic 4, Sec. 7.5).

The radius. From $r^2 = z_1^2 + z_2^2$ the chain rule gives $2r\dot{r} = 2z_1\dot{z}_1 + 2z_2\dot{z}_2$. Substituting $\dot{z}_1 = \rho z_1 + \omega z_2$ and $\dot{z}_2 = -\omega z_1 + \rho z_2$:

$$r\dot{r} = z_1(\rho z_1 + \omega z_2) + z_2(-\omega z_1 + \rho z_2) = \rho z_1^2 + \omega z_1 z_2 - \omega z_1 z_2 + \rho z_2^2 = \rho r^2$$

The terms in ω cancel. Dividing by $r \neq 0$ (the origin is the equilibrium): $\dot{r} = \rho r$.

The angle. Differentiate $\tan\theta = z_2/z_1$ with the chain rule on the left and the quotient rule on the right: $\sec^2\theta\,\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/z_1^2$. Since $\sec^2\theta = r^2/z_1^2$, $\dot{\theta} = (z_1\dot{z}_2 - z_2\dot{z}_1)/r^2$, and

$$z_1\dot z_2 - z_2\dot z_1 = z_1(-\omega z_1 + \rho z_2) - z_2(\rho z_1 + \omega z_2) = -\omega z_1^2 + \rho z_1 z_2 - \rho z_1 z_2 - \omega z_2^2 = -\omega r^2 \implies \dot\theta = -\omega$$

Integration. $\frac{d}{dt}\left(e^{-\rho t} r\right) = e^{-\rho t}(\dot r - \rho r) = 0$, and Barrow's rule between 0 and t, $\left[e^{-\rho s} r(s)\right]_0^t = 0$, gives $r(t) = r_0 e^{\rho t}$. For the angle, $\theta(t) - \theta_0 = \int_0^t (-\omega)\, ds = -\omega t$. Back to Cartesian coordinates, with $c_1 = r_0\cos\theta_0 = z_1(0)$, $c_2 = r_0\sin\theta_0 = z_2(0)$ and the subtraction formulas $\cos(\theta_0 - \omega t) = \cos\theta_0\cos\omega t + \sin\theta_0\sin\omega t$, $\sin(\theta_0 - \omega t) = \sin\theta_0\cos\omega t - \cos\theta_0\sin\omega t$:

$$z_1(t) = e^{-13t}\left(c_1\cos 17t + c_2\sin 17t\right), \qquad z_2(t) = e^{-13t}\left(c_2\cos 17t - c_1\sin 17t\right)$$

Check by substitution (product rule and chain rule): $\dot z_1 = \rho z_1 + e^{\rho t}\omega\left(-c_1\sin\omega t + c_2\cos\omega t\right) = \rho z_1 + \omega z_2$ and $\dot z_2 = \rho z_2 + e^{\rho t}\omega\left(-c_2\sin\omega t - c_1\cos\omega t\right) = \rho z_2 - \omega z_1$, as required.

Back to the original coordinates

$\mathbf{x} = Q\mathbf{z} = z_1\mathbf{u} + z_2\mathbf{w}$, so

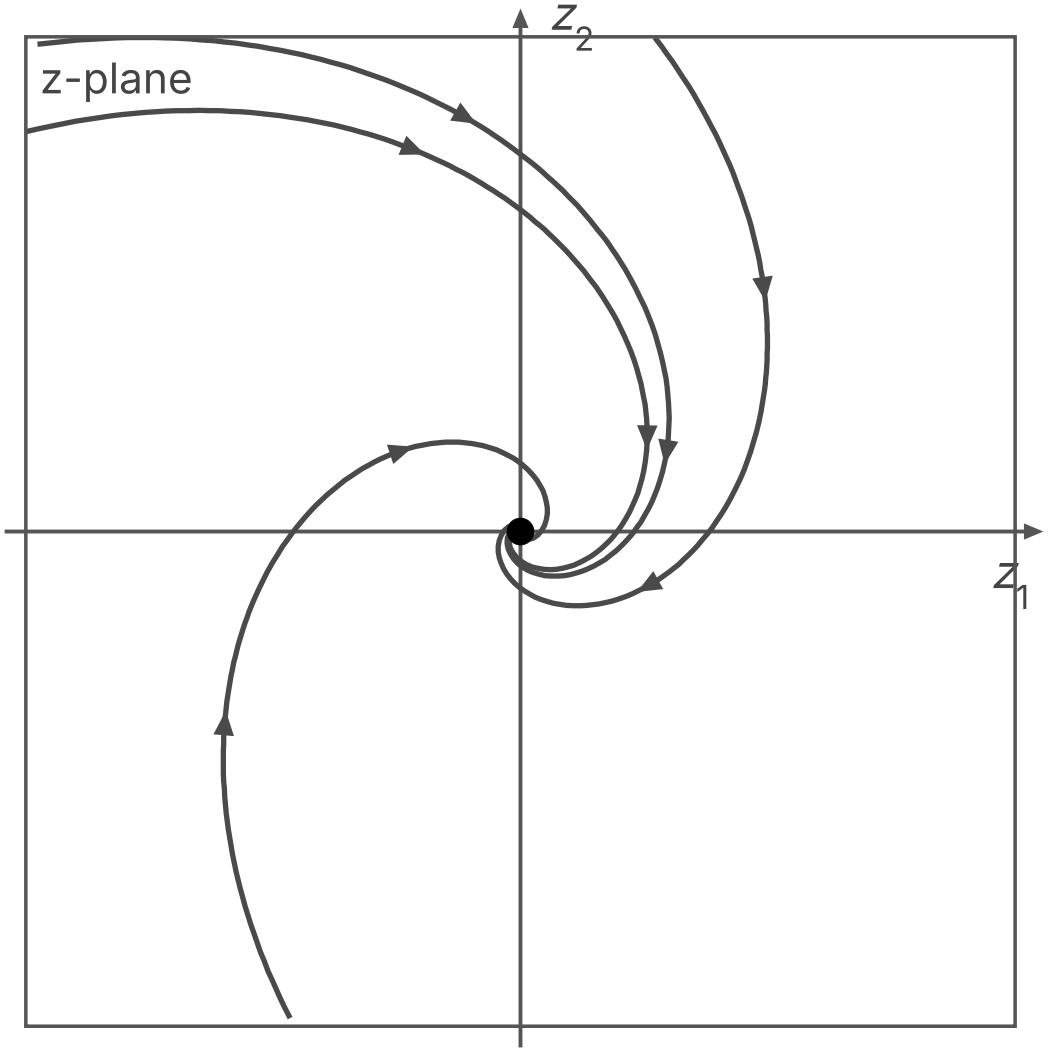
$$\mathbf{x}(t) = e^{-13t}\left[\left(c_1\cos 17t + c_2\sin 17t\right)\begin{pmatrix}-8\\4\end{pmatrix} + \left(c_2\cos 17t - c_1\sin 17t\right)\begin{pmatrix}6\\-5\end{pmatrix}\right]$$

$$x(t) = -6c_1e^{-13t}\sin 17t - 8c_1e^{-13t}\cos 17t - 8c_2e^{-13t}\sin 17t + 6c_2e^{-13t}\cos 17t, \qquad y(t) = 5c_1e^{-13t}\sin 17t + 4c_1e^{-13t}\cos 17t + 4c_2e^{-13t}\sin 17t - 5c_2e^{-13t}\cos 17t$$

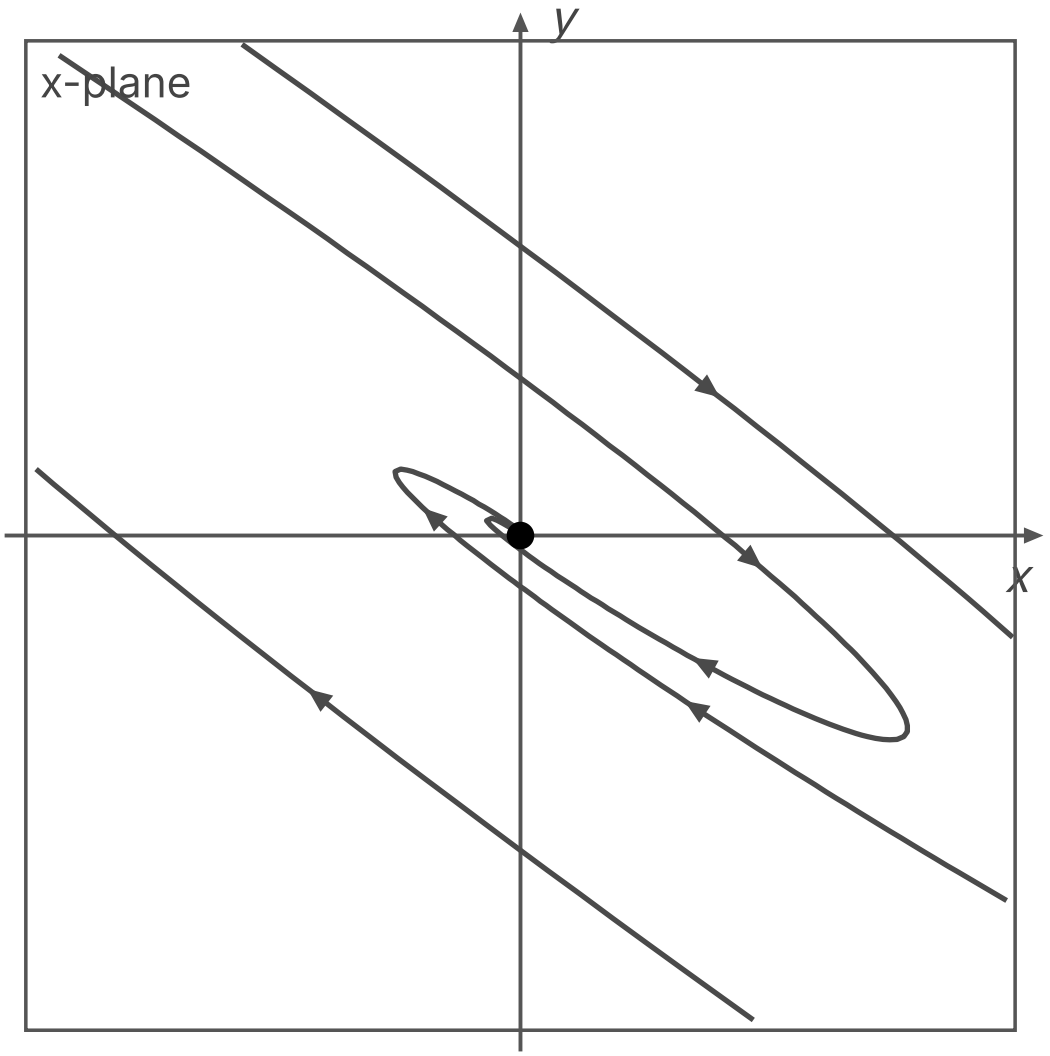
Phase portrait

$\rho = -13$ and $\omega = 17$: the origin is a **stable spiral**. In the $\mathbf{z}$-plane $r = r_0 e^{-13t}$ and $\theta = \theta_0 - 17t$: the motion is always **clockwise** ($\dot\theta = -\omega < 0$), and one full turn takes $T = 2\pi/\omega$ while the radius is multiplied by $e^{2\pi\rho/\omega} = e^{-\frac{26\pi}{17}}$ (shrinking).

Sense of rotation in the $\mathbf{x}$-plane, tested directly on the vector field (Topic 4 Sec. 7.8): at $(0,1)^T$, $\dot{\mathbf{x}} = A(0,1)^T = \begin{pmatrix}\frac{425}{4}\\ -\frac{631}{8}\end{pmatrix}$, so $\dot x = \frac{425}{4}$ is > 0: to the right at the top of the plane: the rotation is **clockwise**. Cross-check: $\det Q = 16$ is > 0, so Q preserves orientation and the clockwise motion of the $\mathbf{z}$-plane becomes clockwise in the $\mathbf{x}$-plane, in agreement.



Canonical $\mathbf{z}$-plane (stable spiral): clockwise, $\dot{\theta} = -\omega$.



Original (x, y)-plane (stable spiral), clockwise.

PHASE 4: INTERPRETATION AND CHECKS

Item	A	Type	Sense of rotation
(i)	$\begin{pmatrix} 7 & -3 \\ 15 & -5 \end{pmatrix}$	unstable spiral	counter-clockwise
(ii)	$\begin{pmatrix} -3 & -3 \\ 6 & 3 \end{pmatrix}$	centre	counter-clockwise
(iii)	$\begin{pmatrix} -\frac{5}{2} & \frac{1}{2} \\ -\frac{5}{2} & -\frac{3}{2} \end{pmatrix}$	stable spiral	clockwise
(iv)	$\begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$	stable spiral	clockwise
(v)	$\begin{pmatrix} \frac{2}{5} & -\frac{6}{5} \\ \frac{82}{15} & \frac{18}{5} \end{pmatrix}$	unstable spiral	counter-clockwise
(vi)	$\begin{pmatrix} -\frac{5}{2} & -\frac{87}{2} \\ \frac{3}{2} & \frac{25}{2} \end{pmatrix}$	unstable spiral	counter-clockwise

Item	A	Type	Sense of rotation
(vii)	$\begin{pmatrix} -\frac{7}{3} & \frac{14}{3} \\ -\frac{35}{3} & \frac{7}{3} \end{pmatrix}$	centre	clockwise
(viii)	$\begin{pmatrix} \frac{423}{8} & \frac{425}{4} \\ -\frac{697}{16} & -\frac{631}{8} \end{pmatrix}$	stable spiral	clockwise

- In every item $\det Q$ and the sign of a_{12} agree (the code asserts $\det Q > 0 \Leftrightarrow a_{12} > 0$), which is the Topic 4 rule: the $\mathbf{z}$-plane turns clockwise, and Q reverses the sense exactly when $\det Q < 0$.
- Centres (ii) and (vii) ($\rho = 0$): $\operatorname{tr} A = 0$ and $\det A = \omega^2 > 0$; the orbits are closed ellipses with period $2\pi/\omega$ ($2\pi/3$ and $2\pi/7$).
- Spirals: (i), (v), (vi) unstable; (iii), (iv), (viii) stable. The growth/decay factor per turn is $e^{2\pi\rho/\omega}$, stated in each item; (viii) decays extremely fast ($e^{-26\pi/17} \approx 0.008$ per turn), so its spiral is nearly invisible after one turn.
- The eigen-data are consistent: $A\boldsymbol{\eta}_+ = \lambda_+ \boldsymbol{\eta}_+$ holds in each item by construction of A (Eq. 7.4 verified entry by entry).